

RESEARCH ARTICLE

Functional Ecology



The PhenObs initiative: A standardised protocol for monitoring phenological responses to climate change using herbaceous plant species in botanical gardens

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Abstract

1. Changes in phenology induced by climate change occur across the globe with important implications for ecosystem functioning and services, species performance and trophic interactions. Much of the work on phenology, especially leaf out and flowering, has been conducted on woody plant species. Less is known about the responses in phenology of herbaceous species induced by global change even though they represent a large and important part of biodiversity worldwide. A globally coordinated research effort is needed to understand the drivers and implications of such changes and to predict effects of global change on plant species phenology and related ecosystem processes.
2. Here, we present the rationale of the PhenObs initiative—botanical gardens as a global phenological observation network. The initiative aims to collect data on plant phenology in botanical gardens which will be used alongside information on plant traits and site conditions to answer questions related to the consequences of global change:
 - A What is the variation in plant phenology in herbaceous species across the growing season and in response to changes in climate?
 - B How can plant phenology be predicted from species' trait composition, provenance, position and extent of the distribution range and species' phylogeny?
 - C What are the implications of this variation with respect to species performance and assembly, biotic interactions (e.g. plant–pollinator interactions) as well as ecosystem processes and services under changing land use and climate?

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